

Linear Algebra

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Introduction to matrices and vectors

A student purchases some items for breakfast:

3 buns and 1 liter milk - total price 2.10 €

Next day a friend visits. He purchases

5 buns and 2 liter milk – total price 4.00 €

Task:

Determine the price for 1 bun and 1 liter of milk.

Step 1: Formulate the above statement in mathematical form

$$3 \cdot B + 1 \cdot M = 2.10 \quad 2 \text{ (coupled) equations } \Rightarrow$$

$$5 \cdot B + 2 \cdot M = 4.00 \quad \text{system of equations}$$

B: price for a bun, M: price for 1ℓ milk

Step 2: solve the system of equations

Method 1: Solution by addition

$$\begin{array}{rcl} 3 \cdot B + 1 \cdot M = 2.10 & | \cdot 2 & \\ \hline 5 \cdot B + 2 \cdot M = 4.00 & & \\ \hline 6 \cdot B + 2 \cdot M = 4.20 & | + & \\ 5 \cdot B + 2 \cdot M = 4.00 & | - & \\ \hline 1 \cdot B + 0 \cdot M = 0.20 & & \\ B = 0.20 & & \end{array}$$

Verification:

$$\begin{array}{l} 3 \cdot B + 1 \cdot M = 2.10 \\ \Rightarrow 3 \cdot 0.2 + 1 \cdot 1.5 = 2.10 \\ \Rightarrow 0.6 + 1 \cdot 1.5 = 2.10 \\ \hline 5 \cdot B + 2 \cdot M = 4.00 \\ 5 \cdot 0.2 + 2 \cdot 1.5 = 4 \\ 1 + 3 = 4 \end{array}$$

Insert this result in one of the basic equations (for example the first equation)

$$3 \cdot 0.2 + M = 0.60 + M = 2.10$$

Subtract 0.6 from both sides of the above equation

$$M = -0.60 + 0.60 + M = 2.10 - 0.60 = 1.50$$

Step 2: Solve the system of equations

Method 2: equating identical entities

$$\begin{array}{r} M = 2.10 - 3 \cdot B \\ M = \frac{1}{2} \cdot (4.00 - 5 \cdot B) \\ \hline 2.10 - 3 \cdot B = 2 - 2.5 \cdot B \\ 0.1 = 2.5 \cdot B \\ B = 0.20 \end{array}$$

Verification:

$$\begin{array}{r} M = 2.10 - 3 \cdot B \\ \Rightarrow M = 2.10 - 3 \cdot 0.2 = 1.5 \\ \hline M = \frac{1}{2} \cdot (4.00 - 5 \cdot B) = \frac{1}{2} \cdot (4.00 - 5 \cdot 0.2) \\ M = \frac{1}{2} \cdot 3 = 1.5 \end{array}$$

Insert this result into one of the above equations (for example the first one)

$$M = 2.10 - 3 \cdot B = 2.10 - 3 \cdot 0.2 = 1.50$$

Step 2: Solve the system of equations – formal solution

Write the system of simultaneous equations in the form

$$\begin{aligned} 3 \cdot B + 1 \cdot M &= 2.10 \\ 5 \cdot B + 2 \cdot M &= 4.00 \end{aligned} \quad \begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix} \cdot \begin{pmatrix} B \\ M \end{pmatrix} = \begin{pmatrix} 2.1 \\ 4 \end{pmatrix}$$

Here $\bar{\bar{A}} = \begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix}$ is a 2x2 **Matrix** (read: 2 by 2 matrix)

and $\bar{v} = \begin{pmatrix} B \\ M \end{pmatrix}$ or $\bar{a} = \begin{pmatrix} 2.1 \\ 4 \end{pmatrix}$ are **vectors** (specifically: 2x1 matrices)

The above equation may now be written as a **matrix equation**:

$$\bar{\bar{A}} \cdot \bar{v} = \bar{a}$$

Properties of matrices

A matrix (plural: matrices) is a rectangular arrangement of numbers:

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1j} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2j} & \dots & a_{2n} \\ \dots & & & & & & \\ a_{i1} & a_{i2} & a_{i3} & \dots & a_{ij} & \dots & a_{in} \\ \dots & \dots & \dots & & & & \dots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mj} & \dots & a_{mn} \end{pmatrix}$$

1. index: line (max: m)
2. index: column (max: n)

This matrix is a m x n matrix
(read: m by n)

m and n specify the **order / rank** of the matrix

The element a_{ij} is located at line i
and column j.

a 1 x 1 matrix is a **scalar**

Some special matrices

Null matrix: $\bar{\bar{0}} = 0 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ (this matrix has zeros everywhere)

Unit matrix $\bar{\bar{E}} = 1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ (matrix with 1 on the matrix diagonal)

Unit element

Diagonal matrix: $\bar{\bar{D}} = \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix}$ (matrix with (different) numbers
on matrix diagonal, all others are zero)

Some special matrices

square matrix: $\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & & & \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{pmatrix}$ (identical number of lines
and columns)

triangular matrix: $\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ 0 & a_{22} & \dots & a_{2n} \\ \dots & & \dots & \\ 0 & 0 & \dots & a_{nn} \end{pmatrix}$ (zeros on one side of
diagonal)

Some special matrices

Column vector: $\begin{pmatrix} a_{11} \\ a_{21} \\ \dots \\ a_{n1} \end{pmatrix} = \begin{pmatrix} a_1 \\ a_2 \\ \dots \\ a_n \end{pmatrix}$ (a $n \times 1$ matrix)

Line vector: $(a_{11} \ a_{12} \ \dots \ a_{1n}) = (a_1 \ a_2 \ \dots \ a_n)$
(a $1 \times n$ matrix)

Unit vektor: $(0 \ 0 \ a_i = 1 \ 0)$ (it has only one 1,
otherwise zeros)

Working with matrices (examples with 3x3 Matrices)

Addition / Subtraction

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \pm \begin{pmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{pmatrix} = \begin{pmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{pmatrix}$$

with

$$a_{ij} \pm b_{ij} = c_{ij} \quad i, j \in \{1, 2, 3\}$$

Remark:

All matrices $\bar{A}$, $\bar{B}$ and $\bar{C}$ have the same rank!

Doing calculations with matrices (examples with 3x3 matrices)

For additions and subtractions have:

$$\bar{A} \pm \bar{B} = \pm \bar{B} + \bar{A}$$

Remark:

For addition and subtraction the order of matrices may be changed => this operation is **commutative**.

Calculations with matrices (examples using 3x3 matrices)

Multiplication

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \cdot \begin{pmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{pmatrix} = \begin{pmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{pmatrix} =$$
$$\begin{pmatrix} a_{11} \cdot b_{11} + a_{12} \cdot b_{21} + a_{13} \cdot b_{31} & a_{11} \cdot b_{12} + a_{12} \cdot b_{22} + a_{13} \cdot b_{32} & a_{11} \cdot b_{13} + a_{12} \cdot b_{23} + a_{13} \cdot b_{33} \\ a_{21} \cdot b_{11} + a_{22} \cdot b_{21} + a_{23} \cdot b_{31} & a_{21} \cdot b_{12} + a_{22} \cdot b_{22} + a_{23} \cdot b_{32} & a_{21} \cdot b_{13} + a_{22} \cdot b_{23} + a_{23} \cdot b_{33} \\ a_{31} \cdot b_{11} + a_{32} \cdot b_{21} + a_{33} \cdot b_{31} & a_{31} \cdot b_{12} + a_{32} \cdot b_{22} + a_{33} \cdot b_{32} & a_{31} \cdot b_{13} + a_{12} \cdot b_{33} + a_{33} \cdot b_{33} \end{pmatrix}$$

here

$$c_{ik} = \sum_{j=1}^3 a_{ij} \cdot b_{jk} = a_{i1} \cdot b_{1k} + a_{i2} \cdot b_{2k} + a_{i3} \cdot b_{3k}$$

Remark:

matrices $\bar{A}$ ($n \times m$), $\bar{B}$ ($m \times p$) and $\bar{C}$ ($n \times p$) may possess different rank!

Calculations using matrices (examples with 3x3 matrices)

Transposition

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}^T = \begin{pmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \\ a_{13} & a_{23} & a_{33} \end{pmatrix}$$

Remark:

A transposition will exchange lines and columns (mirror operation across main diagonal)

Example:

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}^T = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}$$

Some special matrices

A symmetric matrix:

$$\begin{pmatrix} a_{11} & a_{21} & a_{31} \\ a_{21} & a_{22} & a_{32} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

symmetric matrices have the property: $\bar{\bar{A}}^T = \bar{\bar{A}}$

Antisymmetric matrix:

$$\begin{pmatrix} 0 & -a_{21} & a_{31} \\ a_{21} & 0 & -a_{32} \\ -a_{31} & a_{32} & 0 \end{pmatrix}$$

antisymmetric matrices have the property: $\bar{\bar{A}}^T = -\bar{\bar{A}}$

note: there are all zeros on the main diagonal

Matrix inversion – example 1

Task: determine the inverse of the matrix $\begin{pmatrix} 1 & 2 & 0 \\ -1 & 0 & -1 \\ 1 & 1 & 0 \end{pmatrix}$

start with matrix

$$\begin{pmatrix} 1 & 2 & 0 \\ -1 & 0 & -1 \\ 1 & 1 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 0 \\ 0 & 2 & -1 \\ 0 & 1 & 0 \end{pmatrix}$$

unit matrix

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & -1 \end{pmatrix}$$

add line 1 and 2 => line 2
line 1 – line 3 => line 3

$$\begin{pmatrix} 1 & 2 & 0 \\ 0 & 2 & -1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & -1 \end{pmatrix}$$

-2*line 3 + line 1 => line 1

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & -1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$\begin{pmatrix} -1 & 0 & 2 \\ 1 & 1 & 0 \\ 1 & 0 & -1 \end{pmatrix}$$

exchange line 2 and 3

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & -1 \end{pmatrix}$$

$$\begin{pmatrix} -1 & 0 & 2 \\ 1 & 0 & -1 \\ 1 & 1 & 0 \end{pmatrix}$$

add -2*line 2 to 3 => line 3

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

$$\begin{pmatrix} -1 & 0 & 2 \\ 1 & 0 & -1 \\ -1 & 1 & 2 \end{pmatrix}$$

multiply line 3 by -1

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

unit matrix

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} -1 & 0 & 2 \\ 1 & 0 & -1 \\ -1 & 1 & 2 \end{pmatrix}$$

Multiply line 3 by -1

inverse matrix

$$\begin{pmatrix} -1 & 0 & 2 \\ 1 & 0 & -1 \\ 1 & -1 & -2 \end{pmatrix}$$

Verification:

$$\begin{pmatrix} -1 & 0 & 2 \\ 1 & 0 & -1 \\ 1 & -1 & -2 \end{pmatrix} \cdot \begin{pmatrix} 1 & 2 & 0 \\ -1 & 0 & -1 \\ 1 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 0 \\ -1 & 0 & -1 \\ 1 & 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} -1 & 0 & 2 \\ 1 & 0 & -1 \\ 1 & -1 & -2 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Special matrices

$$\bar{\bar{E}} = \begin{pmatrix} 1 & 0 & 0 & \dots & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & \dots & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & \dots & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \dots & 0 & 0 & 0 & 0 & 0 \\ \dots & \dots & \dots & \dots & 1_{ii} & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \dots & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & \dots & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & \dots & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

unit matrix:

only 1 on main diagonal, otherwise 0

$$\bar{\bar{T}}_{ij} = \begin{pmatrix} 1 & 0 & 0 & \dots & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & \dots & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & \dots & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \dots & 0 & 0 & 0 & 0 & 0 \\ \dots & \dots & \dots & \dots & 0_{ii} & \dots & 1_{ij} & \dots & \dots \\ 0 & 0 & 0 & \dots & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \dots & 1_{ji} & 0 & 0_{jj} & 0 & 0 \\ 0 & 0 & 0 & \dots & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & \dots & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

Matrix allowing exchange of lines:

1 on main diagonal, at ii and jj have 0

Non-diagonal elements: 0 almost everywhere, but positions ij and ji with 1

$\bar{\bar{T}}_{ij} \cdot \bar{\bar{M}}$ exchanges lines i and j

$\bar{\bar{M}} \cdot \bar{\bar{T}}_{ij}$ exchanges columns i and j

$$\bar{\bar{T}}_i = \begin{pmatrix} 1 & 0 & 0 & \dots & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & \dots & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & \dots & 0 & 0 & 0 & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & a_{ii} & 0 & 0 & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & \dots & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & \dots & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

matrix multiplying a line by a constant:
 On main diagonal 1 almost everywhere,
 at position ii have factor a
 Off-diagonal : everywhere 0,
 $\bar{\bar{T}}_i \cdot \bar{\bar{M}}$ line i is multiplied by a
 $\bar{\bar{M}} \cdot \bar{\bar{T}}_i$ column i is multiplied by a

$$\bar{\bar{T}}_i = \begin{pmatrix} 1 & 0 & 0 & \dots & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & \dots & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & \dots & 0 & 0 & 0 & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & a_{ii} & 0 & 1_{ij} & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & \dots & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & \dots & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

Multiply line i by a and add line j

Determinants

Determinants are introduced as these will allow us to determine whether or not a square matrix has an inverse, i.e. whether or not it is **invertible**.

A determinant is a real number, which is obtained from all entries of the matrix. **Note**, that matrices which are not square do not have a determinant.

Determinants are useful in a number of different contexts, and therefore determinants appear in different contexts in the literature. This may be the reason that a different manner of writing has emerged. However, its relationship to matrix notation has been retained to a large extent.

Matrix

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

Round brackets

Determinant

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

2 parallel lines

The difference in notation is minute, but the calculation instructions are quite different => danger of confusion and misinterpretation

Instructions for the calculation of 2 x 2 Determinants

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = a \cdot d - bc$$

Examples:

$$\begin{vmatrix} 7 & 12 \\ 4 & 7 \end{vmatrix} = 7 \cdot 7 - 4 \cdot 12 = 49 - 48 = 1, \quad \begin{vmatrix} 6 & \frac{1}{6} \\ 0,6 & 1,5 \end{vmatrix} = 6 \cdot 1,5 - 0,6 \cdot \frac{1}{6} = 9 - \frac{1}{10} = 8,9$$

For the 2 x 2 determinant (2 lines and 2 columns)
the following instruction is stated:

„Product of elements on main diagonal
minus product of off-diagonal elements“!

For a matrix $\bar{\bar{A}}$ with

$$\bar{\bar{A}} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

The associated determinant is given as :

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = |\bar{\bar{A}}| = \det(\bar{\bar{A}})$$

„Larger“ Determinants of Type (n x n), n > 2

Determinants exist for any arbitrary square matrix. Now the question arises as to how to calculate this determinant. Here the following process is useful: „develop the determinant (using, for example, the 1st line)“.

This process is introduced using a 3 x 3 matrix:

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11} \cdot \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \cdot \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \cdot \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

The pattern of a chess board is used here for the sign!

Develop determinant using 1st line: cross out 1st line and column of element under consideration, multiply element with smaller determinant => reduced determinant.

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11} \cdot \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \cdot \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \cdot \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

Expanding the determinant using the first line reduces the 3 x 3 determinant to a sequence of 2 x 2 determinants.

We already know how to evaluate the 2 x 2 determinants.

The sign is determined using the chess board distribution of signs.

Let i and j be the indices of the a_{ij} coefficient, then use

(+) if $i+j$ is even

or (-) if $i+j$ is odd.

$$sign = (-1)^{i+j}$$

Example:

$$\begin{vmatrix} 4 & 2 & 0 \\ 1 & 2 & 4 \\ 0 & 1 & 7 \end{vmatrix} = 4 \cdot \begin{vmatrix} 2 & 4 \\ 1 & 7 \end{vmatrix} - 2 \cdot \begin{vmatrix} 1 & 4 \\ 0 & 7 \end{vmatrix} + 0 \cdot \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix} = 4 \cdot 10 - 2 \cdot 7 + 0 = 26$$

This is the principle, with the help of which a larger determinant may be reduced, in several steps if necessary, to a sequence of 2 x 2 determinants. Thus a method is available to reduce larger determinants to smaller one.

However, this method is cumbersome and lengthy if the determinants are of larger size.

The examples of 3×3 matrices indicates that the **appearance of zeros** significantly simplifies calculations!

It is an advantage that some properties of determinants allow the „creation of zeros“ or, in some cases, to even determine the value of the determinant by inspection only.

Properties of Determinants - Property (a)

If two lines are exchanged in a matrix, then the determinant changes sign.

The same is true for the exchange of two columns in the matrix.

Example:

$$\bar{A} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}, \quad \bar{A}_{1 \leftrightarrow 2} = \begin{pmatrix} a_{12} & a_{11} & a_{13} \\ a_{22} & a_{21} & a_{23} \\ a_{32} & a_{31} & a_{33} \end{pmatrix}$$

$$\Rightarrow \det(\bar{A}) = -\det(\bar{A}_{1 \leftrightarrow 2})$$

$$|\bar{\bar{A}}| = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11} \cdot \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \cdot \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \cdot \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} =$$

$$= a_{11} \cdot (a_{22} \cdot a_{33} - a_{32} \cdot a_{23}) - a_{12} \cdot (a_{21} \cdot a_{33} - a_{31} \cdot a_{23}) + a_{13} \cdot (a_{21} \cdot a_{32} - a_{31} \cdot a_{22})$$

$$|\bar{\bar{A}}_{1 \leftrightarrow 2}| = \begin{vmatrix} a_{12} & a_{11} & a_{13} \\ a_{22} & a_{21} & a_{23} \\ a_{32} & a_{31} & a_{33} \end{vmatrix} = a_{12} \cdot \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} - a_{11} \cdot \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} + a_{13} \cdot \begin{vmatrix} a_{22} & a_{21} \\ a_{32} & a_{31} \end{vmatrix} =$$

$$= a_{12} \cdot (a_{21} \cdot a_{33} - a_{31} \cdot a_{23}) - a_{11} \cdot (a_{22} \cdot a_{33} - a_{32} \cdot a_{23}) + a_{13} \cdot (a_{31} \cdot a_{22} - a_{21} \cdot a_{32})$$

$$\Rightarrow \det(\bar{\bar{A}}) = -\det(\bar{\bar{A}}_{1 \leftrightarrow 2})$$

$$|\bar{A}| = \begin{vmatrix} 1 & 6 & 3 \\ 1 & 5 & 4 \\ 0 & 1 & 0 \end{vmatrix} = 1 \cdot \begin{vmatrix} 5 & 4 \\ 1 & 0 \end{vmatrix} - 6 \cdot \begin{vmatrix} 1 & 4 \\ 0 & 0 \end{vmatrix} + 3 \cdot \begin{vmatrix} 1 & 5 \\ 0 & 1 \end{vmatrix} \\ = 1 \cdot (5 \cdot 0 - 1 \cdot 4) - 6 \cdot (1 \cdot 0 - 0 \cdot 4) + 3 \cdot (1 \cdot 1 - 0 \cdot 5) = -4 - 0 + 3 = -1$$

Exchange 1st and 3rd line!

$$|\bar{A}_{1 \leftrightarrow 3}| = \begin{vmatrix} 3 & 6 & 1 \\ 4 & 5 & 1 \\ 0 & 1 & 0 \end{vmatrix} = 3 \cdot \begin{vmatrix} 5 & 1 \\ 1 & 0 \end{vmatrix} - 6 \cdot \begin{vmatrix} 4 & 1 \\ 0 & 0 \end{vmatrix} + 1 \cdot \begin{vmatrix} 4 & 5 \\ 0 & 1 \end{vmatrix} = \\ = 3 \cdot (5 \cdot 0 - 1 \cdot 1) - 6 \cdot (4 \cdot 0 - 0 \cdot 1) + 1 \cdot (4 \cdot 1 - 0 \cdot 5) = -3 - 0 + 4 = +1$$

The above property can be used to simplify the expression for the determinant, and to avoid pitfalls of calculation.

Property (b)

Given a matrix $\bar{H}$ with $\det(\bar{H}) = |\bar{H}|$. Let the entries of one particular line (or column) be a multiple of other number with the multiplying factor being λ . If this factor is taken out of the determinant, then the remaining part of $|\bar{H}|$ may be easier to handle and calculate. Then have $|H| = \lambda \cdot |\hat{H}|$, where $\hat{H}$ is the matrix with the elements of the line (or column) divided by λ .

Example:

$$|H| = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ \lambda \cdot a_{31} & \lambda \cdot a_{32} & \lambda \cdot a_{33} \end{vmatrix} = \lambda \cdot \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \lambda \cdot |\hat{H}|$$

$$\begin{aligned}
 |H| &= \begin{vmatrix} \frac{1}{3} & \frac{2}{3} & 1 \\ 1 & \frac{2}{5} & 4 \\ \frac{3}{7} & 0 & \frac{8}{7} \end{vmatrix} = \frac{1}{3} \cdot \begin{vmatrix} \frac{2}{5} & 4 \\ 0 & \frac{8}{7} \end{vmatrix} - \frac{2}{3} \cdot \begin{vmatrix} 1 & 4 \\ \frac{3}{7} & \frac{8}{7} \end{vmatrix} + 1 \cdot \begin{vmatrix} 1 & \frac{2}{5} \\ \frac{3}{7} & 0 \end{vmatrix} = \\
 &= \frac{1}{3} \cdot \left(\frac{16}{35} - 0 \right) - \frac{2}{3} \cdot \left(\frac{8}{7} - \frac{12}{7} \right) + 1 \cdot \left(0 - \frac{6}{35} \right) = \frac{16}{105} + \frac{8}{21} - \frac{6}{35} = \frac{16+40-18}{105} \\
 &= \frac{38}{105}
 \end{aligned}$$

Alternative calculation:

$$\begin{aligned}
 \Rightarrow \begin{vmatrix} \frac{1}{3} & \frac{2}{3} & 1 \\ 1 & \frac{2}{5} & 4 \\ \frac{3}{7} & 0 & \frac{8}{7} \end{vmatrix} &= \frac{1}{3} \cdot \begin{vmatrix} 1 & 2 & 3 \\ 1 & \frac{2}{5} & 4 \\ \frac{3}{7} & 0 & \frac{8}{7} \end{vmatrix} = \frac{1}{3} \cdot \frac{1}{7} \cdot \begin{vmatrix} 1 & 2 & 3 \\ 1 & \frac{2}{5} & 4 \\ 3 & 0 & 8 \end{vmatrix} = \frac{1}{3} \cdot \frac{1}{5} \cdot \frac{1}{7} \cdot \begin{vmatrix} 1 & 2 & 3 \\ 5 & 2 & 20 \\ 3 & 0 & 8 \end{vmatrix} \Rightarrow
 \end{aligned}$$

$$|H| = \lambda |\hat{H}| = \frac{1}{105} \cdot \begin{vmatrix} 1 & 2 & 3 \\ 5 & 2 & 20 \\ 3 & 0 & 8 \end{vmatrix} \text{ with } |\hat{H}| = \begin{vmatrix} 1 & 2 & 3 \\ 5 & 2 & 20 \\ 3 & 0 & 8 \end{vmatrix}$$

$$|\hat{H}| = \begin{vmatrix} 1 & 2 & 3 \\ 5 & 2 & 20 \\ 3 & 0 & 8 \end{vmatrix} = 1 \cdot \begin{vmatrix} 2 & 20 \\ 0 & 8 \end{vmatrix} - 2 \cdot \begin{vmatrix} 5 & 20 \\ 3 & 8 \end{vmatrix} + 3 \cdot \begin{vmatrix} 5 & 2 \\ 3 & 0 \end{vmatrix} =$$

$$= 16 + 40 - 18 = 38$$

Thus one obtains

$$|H| = \lambda |\hat{H}| = \frac{1}{105} \cdot 38 = \frac{38}{105}$$

It is easier to evaluate the determinant of a matrix for which all entries are natural numbers

Property (c)

If one adds a multiple of one line (or column) to another line (or column) then the determinant does not change its value.

Example:

$$\det(\bar{A}) = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}, \quad \det(\bar{A}_{\text{add}}) = \begin{vmatrix} a_{11} + \lambda a_{31} & a_{12} + \lambda a_{32} & a_{13} + \lambda a_{33} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$\Rightarrow \det(\bar{A}) = \det(\bar{A}_{\text{add}})$$

This rule is of importance for practical calculations. Using it one may **generate zeros!**

Example:

$$\begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 1 & 2 \end{vmatrix} = 1 \cdot \begin{vmatrix} 2 & 2 \\ 1 & 2 \end{vmatrix} - 1 \cdot \begin{vmatrix} 1 & 2 \\ 1 & 2 \end{vmatrix} + 1 \cdot \begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix} \\ = (4 - 2) - (2 - 2) + (1 - 2) = 2 - 0 - 1 = 1$$

Subtract line 3 from line 1 and store in line 1!

$$\begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 1 & 2 \end{vmatrix} = \begin{vmatrix} 0 & 0 & -1 \\ 1 & 2 & 2 \\ 1 & 1 & 2 \end{vmatrix} = -1 \cdot \begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix} = -1 + 2 = 1$$

Special Determinants

Now some particular cases are considered. Three special cases are listed for determinants which are zero. In most cases this is easy to spot.

Use the principle of development of a determinant to verify the statement.

- a) Given a matrix for which one line (or column) has only 0 as an entry.
The determinant of this matrix is zero.
- b) Consider a matrix for which two lines (or columns) are identical.
The determinant of this matrix is zero.
- (c) A determinant is 0 if, after taking a factor λ out of a line or column, case b is generated.

As the unit matrix $\bar{\bar{E}}$ is the generalisation of the number 1 to matrix calculus it is reasonable to assume that

$$\det(\bar{\bar{E}}) = 1$$

This assumption is correct.

Example: Develop using the 1st line (repeatedly!).

$$\begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix} = 1 \cdot \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 1 \cdot 1 \cdot \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1 \cdot 1 \cdot 1 \cdot 1$$

2 important mathematical statements

The topics are:

The determinant of a product of matrices and the
Condition for the existence of the inverse matrix.

Both these statements are connected to some fundamental aspects of linear algebra. Their derivation is lengthy and complex. These theorems are stated here without any mathematical proof or derivation.

The determinant of a product of matrices is equal to the product of the determinants of the matrices.

Example:

$$\det(\bar{\bar{A}} \cdot \bar{\bar{B}}) = \det(\bar{\bar{A}}) \cdot \det(\bar{\bar{B}})$$

$$\text{Let } \det(\bar{\bar{A}}) = \begin{vmatrix} 1 & 2 \\ 5 & 3 \end{vmatrix} \text{ and } \det(\bar{\bar{B}}) = \begin{vmatrix} 3 & 2 \\ -4 & 1 \end{vmatrix}$$

$$\text{Then } \det(\bar{\bar{A}} \cdot \bar{\bar{B}}) = \det\left(\begin{pmatrix} 1 & 2 \\ 5 & 3 \end{pmatrix} \cdot \begin{pmatrix} 3 & 2 \\ -4 & 1 \end{pmatrix}\right) = \begin{vmatrix} -5 & 4 \\ 3 & 13 \end{vmatrix} = -77$$

$$\det(\bar{\bar{A}}) = \begin{vmatrix} 1 & 2 \\ 5 & 3 \end{vmatrix} = -7 \quad \text{and} \quad \det(\bar{\bar{B}}) = \begin{vmatrix} 3 & 2 \\ -4 & 1 \end{vmatrix} = +11$$

$$\text{Therefore: } \det(\bar{\bar{A}}) \cdot \det(\bar{\bar{B}}) = -7 \cdot 11 = -77$$

Matrix $\bar{\bar{A}}$ can be inverted if and only if $\det(\bar{\bar{A}}) \neq 0$ gilt.

1st test using an example:

Given $\bar{\bar{A}} = \begin{pmatrix} 6 & 2 \\ 4 & 3 \end{pmatrix}$ has $\det(\bar{\bar{A}}) = 18 - 8 = 10$ with $\bar{\bar{A}}^{-1} = \frac{1}{10} \begin{pmatrix} 3 & -2 \\ -4 & 6 \end{pmatrix}$

Note: $\det(\bar{\bar{A}}) = 10 \neq 0$

Check:

$$\bar{\bar{A}} \cdot \bar{\bar{A}}^{-1} = \frac{1}{10} \begin{pmatrix} 6 & 2 \\ 4 & 3 \end{pmatrix} \cdot \begin{pmatrix} 3 & -2 \\ -4 & 6 \end{pmatrix} = \frac{1}{10} \cdot \begin{pmatrix} 10 & 0 \\ 0 & 10 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

2nd test using an example:

Given a nilpotent matrix $\bar{\bar{A}}_{nilpotent} = \begin{pmatrix} -3 & -9 \\ 1 & 3 \end{pmatrix}$

A matrix is nilpotent if there is a k for which: $(\bar{\bar{A}}_{nilpotent})^k = 0$

Check: $(\bar{\bar{A}}_{nilpotent})^2 = \begin{pmatrix} -3 & -9 \\ 1 & 3 \end{pmatrix} \cdot \begin{pmatrix} -3 & -9 \\ 1 & 3 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$

Evaluate $\det(\bar{\bar{A}}_{nilpotent}) = -9 + 9 = 0$

The inverse matrix of $\bar{\bar{A}}_{nilpotent}$ does not exist.

An Application

Note that the matrix equation $\bar{\bar{A}} \cdot \bar{\bar{A}}^{-1} = 1 = \bar{\bar{E}}_n$
(here $\bar{\bar{E}}_n$ is the $n \times n$ unit matrix) implies that

$$\det(\bar{\bar{A}}) \cdot \det(\bar{\bar{A}}^{-1}) = 1$$

and that therefore $\det(\bar{\bar{A}}^{-1}) = \frac{1}{\det(\bar{\bar{A}})}$

In particular as $\det(\bar{\bar{A}}^{-1}) \neq 0$ the matrix $\bar{\bar{A}}^{-1}$ has an inverse,
namely $(\bar{\bar{A}}^{-1})^{-1} = \bar{\bar{A}}$.